
SaaS Lead Generation & Revenue Intelligence Pipeline

BILL SWIFT

An end-to-end automation project for Bill Swift, a usage-based billing SaaS company

A portfolio project by Praveen Sambasivam

What is Bill Swift?

Bill Swift is a fictional SaaS company built as the business scenario for this portfolio project — modelled on Chargebee, a real subscription billing and revenue management platform used by SaaS companies. Bill Swift gives every part of this pipeline a reason to exist: a webinar, a scored lead, a CRM record, and a revenue forecast that a real billing-SaaS company's sales and finance teams would actually use.

What is this project about?

This project is an automated, end-to-end SaaS lead-generation and revenue-intelligence pipeline. From the moment someone registers for a webinar to the moment their deal closes, every step — scoring, CRM sync, email nurture, revenue prediction, and reporting — runs without manual intervention.

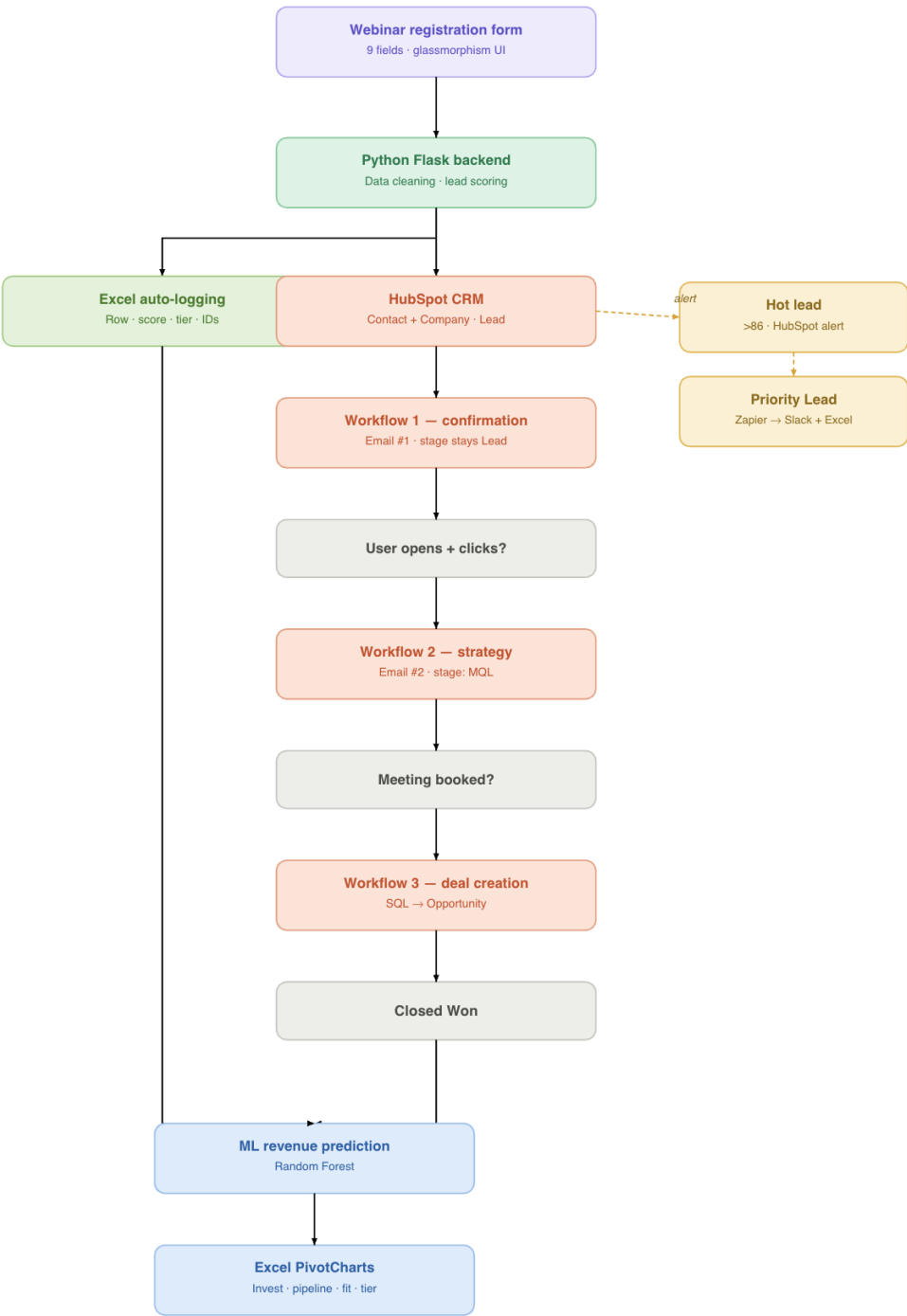
What problem does this project solve?

Every SaaS company faces the same gap between a new lead and a closed deal — leads come in faster than anyone can manually qualify, track, or follow up with. For Bill Swift, I built a system to close that entire gap automatically:

- **Lead qualification** — a 6-signal, 100-point scoring engine scores every lead automatically based on the details captured at signup (company size, transaction volume, current billing setup, and more), so sales isn't guessing who to prioritize.
- **CRM and data consistency** — every lead syncs to Excel and HubSpot simultaneously with matching IDs, keeping one reliable source of truth instead of two drifting records.
- **Hot lead response time** — an automated alert flags any lead scoring above 85, so high-value prospects aren't missed while sitting in a queue.
- **Structured follow-up** — tiered nurture workflows move leads through Lead → MQL → SQL → Opportunity → Closed Won without manual handoffs.
- **Real-time visibility beyond the CRM** — a Zapier webhook posts an instant Slack alert the moment a lead crosses the priority threshold and logs it to a dedicated priority-leads Excel file, so high-value prospects are seen immediately and tracked separately — not just sitting inside HubSpot.
- **Revenue visibility** — an ML revenue-prediction model feeds CXO-ready PivotCharts, showing leadership where the pipeline is strongest and where to invest.

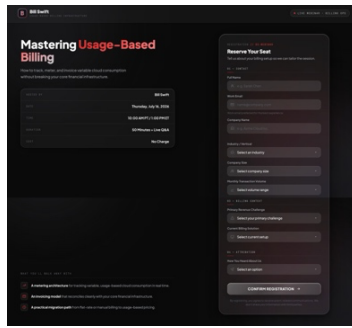
Pipeline Overview

Here's how a lead moves through the system, end to end:

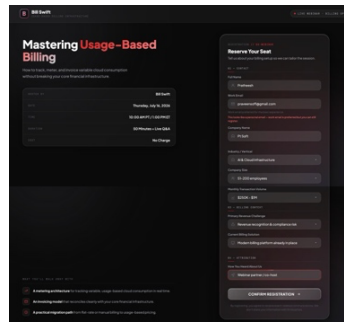


1. Webinar registration form

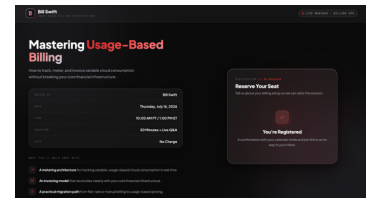
Lead-gen entry point — a webinar signup form that captures contact, company, and billing context in one step.



Full webpage — event details and registration form



Form mid-fill — personal email soft warning shown



Success screen — confirms the pipeline is live

2. Python Flask backend

Receives the form data, cleans it, and triggers scoring — all in one request.

3. Lead scoring engine

Scores each lead out of 100 across 6 signals; sorts into Hot / Warm / Cold.

Signal	Max
Monthly Transaction Volume	28
Primary Revenue Challenge	24
Current Billing Solution	19
Company Size	14
How You Heard About Us	10
Email Type	5

Tiers: 0-35 Cold · 36-65 Warm · 66-100 Hot

Six weighted scoring signals feed a 100-point scale, split into three tiers.

4. Excel auto-logging

Auto-appends every lead to Excel with score, tier, and IDs — no manual entry.

Timestamp	Name	Email	Company	Industry	Company Size	Monthly Transaction Volume	Primary Revenue Challenge	Current Billing Solution	How You Heard About Us	Lead Score	Lead Tier	HubSpot ID	HubSpot ID	Predict
2026-07-01	Praveen Sai	praveen.dur@Prvn Tech	Personal Er	edtech	201-1000	250k-1m	manual-inv	stripe-payp	google	58	Warm	51244505	53152383	152837.3
2026-07-01	Sarah Chen	sarah@veri Vertex AI	vertexai.io	ecommerce	201-1000	50k-250k	usage-mete	in-house	newsletter	65	Warm	51248541	53157054	227258
2026-07-01	David Ross	dross@saa: Saasflow	saasflow.cc	proptech-iot	201-1000	250k-1m	usage-mete	in-house	partner	75	Hot	51244342	53153833	271457.6
2026-07-01	Elena Rostc	elena@cyb Cybercore	cybercore.ei	fintech	11-50	10m+	research	stripe-payp	google	60	Warm	51245502	53149321	184046.1
2026-07-01	Elena Rostc	elena@cyb Cybercore	cybercore.ei	ai-cloud	11-50	250k-1m	research	legacy-ente	referral	45	Warm	51245502	53149321	126484.1
2026-07-01	Marcus Var	marcus@p Paybolt	paybolt.fin	ecommerce	51-200	1m-10m	research	legacy-ente	newsletter	51	Warm	51238017	53154755	157615.1
2026-07-01	Chloe Tana	c.tanaka@Edulearn	edulearn.ci	edtech	1-10	under-50k	rev-recogni	in-house	partner	47	Warm	51235872	53147174	180323.1
2026-07-01	Liam O'Con	liam@strei Streamvib	streamvibe	digital-media	1-10	250k-1m	usage-mete	none-man	other	66	Hot	51242147	53154585	238010.9
2026-07-01	Amara Okal	a.okafor@r Nexthome	nexthome.	proptech-iot	51-200	10m+	research	stripe-payp	linkedin	65	Warm	51235194	53153292	193846.6
2026-07-01	Jacob Mille	j.miller@bi Biotrack	biotrack.or	ai-cloud	1000+	50k-250k	manual-inv	none-man	google	62	Warm	51244507	53149501	171892.2
2026-07-01	Pratheesh	praveensof Pt Soft	Personal Er	ai-cloud	51-200	250k-1m	rev-recogni	modern-pl	partner	49	Warm	51238751	53152383	189583.2

Every submission appends a new row — score, tier, and HubSpot IDs included.

Creates Contact + Company in HubSpot; handles duplicate conflicts automatically.

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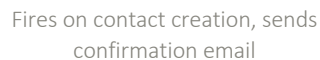
HubSpot Contacts — auto-created on every submission

HubSpot Companies — linked to matching contacts

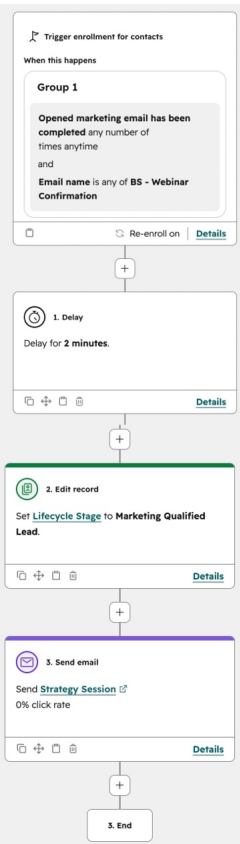
4 workflows: confirmation email → MQL upgrade → SQL/deal creation → hot lead alert.

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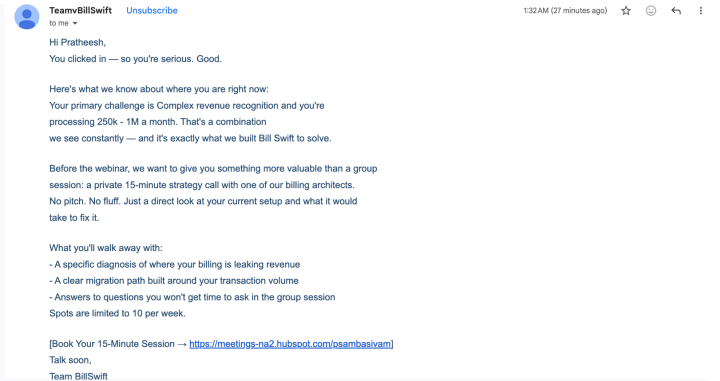
Workflow 1 — registration confirmation



Workflow 2 — MQL upgrade & strategy session

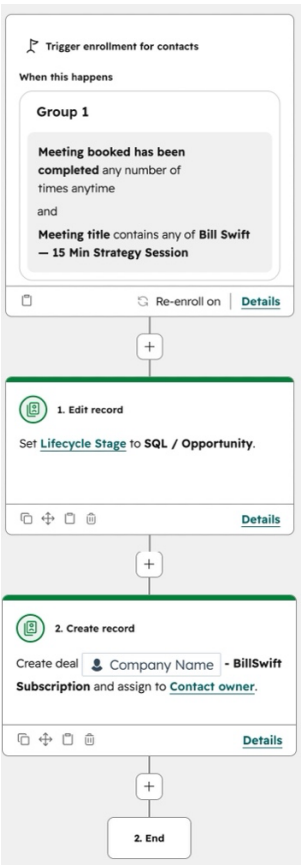


Email open triggers MQL upgrade,
sends Email #2



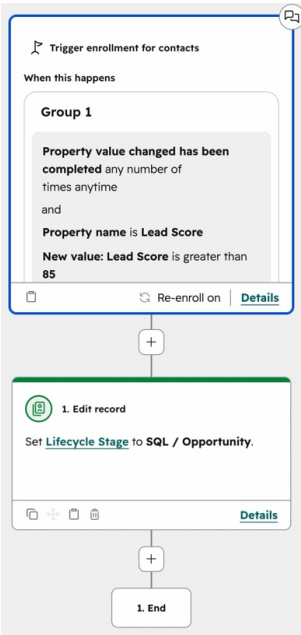
Email #2 — dynamic tokens populate real challenge + volume

Workflow 3 — meeting booked → SQL & deal creation



Meeting booked triggers SQL/Opportunity and auto-creates a deal.

Workflow 4 — hot lead alert (Lead Score > 85)



Lead Score above 85 triggers an instant SQL upgrade and sales alert — independent of the email sequence.

Meeting booked → deal auto-created in pipeline → progresses to Closed Won.



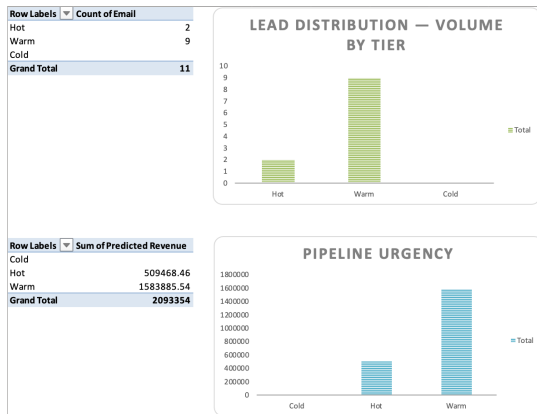
Random Forest model predicts contract value; written back to Excel.

200-row historical dataset used to train the Random Forest model at server startup.

Predicted revenue is written back into the same Excel row as the lead score.

9. Excel PivotCharts

4 live charts: revenue by industry, lead volume, tier distribution, pipeline urgency.



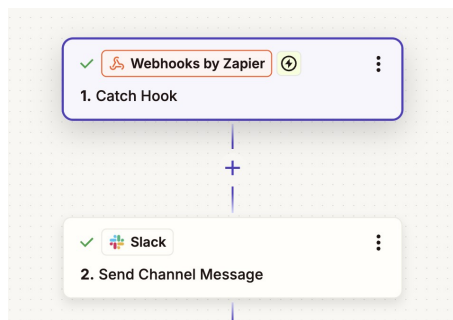
Lead Distribution by Tier (2 Hot · 9 Warm) and Pipeline Urgency



Where to Invest — revenue by industry, and Market Reach by volume

10. Priority Lead Alert (Zapier + Slack)

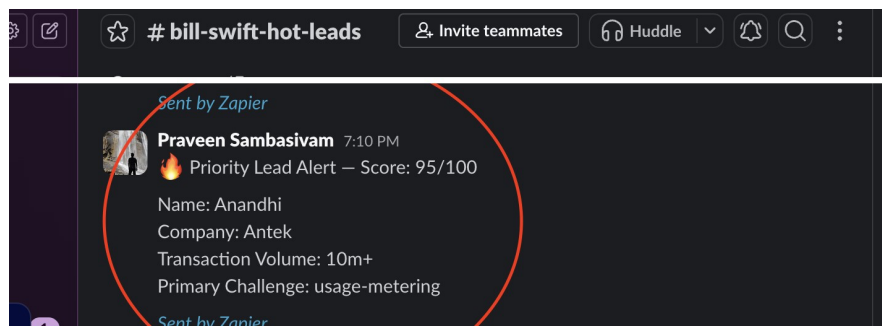
Leads scoring above 85 — a separate, higher bar than the general Hot tier (66+) — trigger a dedicated Zapier webhook, called directly from the Flask backend. This posts a real-time alert to a Slack channel and logs the lead to a separate priority-leads Excel file, extending automation beyond HubSpot's native capabilities to a tool it doesn't natively integrate with.



A two-step Zap: Webhooks by Zapier (Catch Hook) → Slack (Send Channel Message)



Priority lead auto-logged to a separate Excel file, with HubSpot IDs included



Real-time Slack alert — fired the moment a lead crosses the Priority threshold, with score, company, and challenge context

Project Summary

- Zero manual steps — from form submission to CRM contact, ML prediction, and email delivery, everything is automated.
- Server-side lead scoring — scores are calculated in Python and cannot be tampered with from the browser.
- Dynamic email personalization — Email #2 injects real CRM properties into each send.
- Full audit trail — every row contains a timestamp, cleaned data, lead score, tier, HubSpot IDs, and predicted revenue.
- Priority Lead Alert — leads scoring above 85 trigger a Zapier webhook that posts an instant Slack alert and logs the lead to a dedicated priority-leads Excel file, independent of the general lead log.